

Information Retrieval

Assignment Report

This assignment implements and compares six retrieval models on the Cranfield dataset, which contains 1,400 scientific abstracts as documents and 225 search topics as queries. The models evaluated in this assignment are Okapi TF, TF-IDF, BM25, Laplace Language Model, Jelinek-Mercer Language Model, and Elasticsearch built-in BM25. All six models were implemented from scratch using term frequency (TF) and document frequency (DF) statistics retrieved from Elasticsearch. The implementation process involved parsing XML documents, indexing them into Elasticsearch, extracting queries from XML files, building helper functions to retrieve TF, DF, document length, average document length, total document count, and vocabulary size, and then applying each ranking formula to score all 1,400 documents for each of the 225 queries. After scoring, the top 100 results per query were saved in TREC format. Finally, trec_eval was used to compute Mean Average Precision (MAP) scores for each model, and manual comparison tables were created for 5 sample queries showing the top 10 documents returned by ES Built-in, TFIDF, BM25, and Jelinek-Mercer LM. MAP Scores Results

Model	MAP Score
TF-IDF	0.0086
Okapi TF	0.0082
BM25 (Custom)	0.0080
ES Built-in BM25	0.0063
Jelinek-Mercer LM	0.0063
Laplace LM	0.0055

TF-IDF achieved the highest MAP score (0.0086). Laplace LM had the lowest MAP score (0.0055). The custom BM25 implementation (0.0080) outperformed Elasticsearch's built-in BM25 (0.0063).

Table for Query 1

Ran k	ES Built-in	TFID F	BM2 5	Okapi TF	Lapla ce	JelinekMer cer
1	184	184	184	184	1268	1268
2	486	13	486	12	792	184
3	13	486	13	13	14	486
4	1268	12	126 8	51	1147	14
5	12	1268	12	1268	576	51
6	51	51	51	486	51	588
7	878	875	14	14	184	172
8	14	878	878	588	1144	311
9	792	746	875	747	588	792
10	875	14	792	878	1313	576

Table for Query 10

Ran k	ES Built-in	TFID F	BM2 5	Okapi TF	Lapla ce	JelinekMer cer
1	491	491	491	491	315	257
2	257	257	257	257	798	315
3	315	315	315	544	329	344
4	121	386	121	315	522	491
5	148	385	148	1074	257	329
6	767	148	251	558	344	121
7	251	121	767	294	294	610
8	651	1273	651	767	1313	294

9	344	767	344	1302	225	1302
10	296	251	296	651	928	651

Table for Query 50

Ran k	ES Built-in	TFID F	BM2 5	Okapi TF	Lapla ce	JelinekMer cer
1	215	215	215	121	798	121
2	121	277	121	97	1313	212
3	277	121	277	511	329	511
4	511	511	511	1222	927	97
5	216	214	216	277	97	1222
6	212	134	212	186	212	186
7	214	216	214	798	277	213
8	134	212	134	215	89	455
9	588	989	588	212	1239	798
10	945	945	989	213	417	216

Table for Query 15

Ran k	ES Built-in	TFID F	BM2 5	Okapi TF	Lapla ce	JelinekMer cer
1	45	21	45	21	662	45
2	21	45	21	45	45	270
3	550	550	550	550	1204	21
4	22	22	22	22	270	22
5	270	306	270	303	571	550
6	571	571	306	306	1040	303
7	306	270	571	270	1107	306
8	102	102	102	269	564	571

9	1215	1215	1215	398	294	1264
10	303	303	303	571	550	1204

Table for Query 100

Ran k	ES Built-in	TFID F	BM2 5	Okapi TF	Lapla ce	JelinekMer cer
1	292	292	292	292	292	292
2	921	657	921	458	272	458
3	657	912	657	912	458	912
4	912	921	912	381	798	381
5	610	458	610	563	364	610
6	458	610	458	610	381	364
7	364	364	364	661	563	646
8	754	754	754	272	94	962
9	801	661	801	364	89	1268
10	381	381	786	375	962	1072

Conclusion

TF-IDF performed best because IDF penalizes common words and gives more weight to rare, informative terms. In the Cranfield dataset, specialized aerospace terminology appears rarely, making those terms highly discriminative.

Laplace LM performed worst because add-one smoothing adds 1 to every term frequency, including zero counts. With a vocabulary of approximately 247,000 terms, this creates a very flat probability distribution where unseen terms get artificially high probability, diluting the contribution of actual matching terms.

The custom BM25 implementation outperformed Elasticsearch's built-in BM25, suggesting parameter tuning ($k_1=1.2$, $b=0.75$) can significantly impact results compared to default settings.

Language models with aggressive smoothing (Laplace) are less effective for small specialized collections, while vector space models with IDF weighting (TF-IDF and BM25) are more suitable for the Cranfield dataset.